

## SEBES ALBA, Romania

WMO: 152310

Lat: 45.9643N

Lon: 23.5334E

Elev: 272

StdP: 98.10

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 1             | -13.1      | -10.2 | -15.1                         | 1.0 | -12.6 | -12.2 | 1.4 | -8.6 | 12.3                  | -6.3 | 10.5 | -1.7 | 1.2                   | 130  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 13.1                   | 32.5            | 20.3 | 30.7 | 20.1 | 29.0 | 19.6 | 22.1                | 29.0 | 21.3 | 28.1 | 20.6 | 27.1 | 2.2                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.0                            | 15.2 | 24.6 | 19.2 | 14.4 | 23.8 | 18.5 | 13.8 | 23.1 | 66.1          | 29.0 | 63.1 | 27.5 | 60.5 | 27.4 | 25.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 10.1          | 8.6  | 7.2 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.8   | -1.7  | 1.9   | 6.0   | 11.0  | 16.3  | 19.8  | 20.9  | 21.4  | 16.8  | 10.9  | 5.5   | 0.7  |
|                                                                                  | DBStd       | 8.81   | 4.76  | 4.59  | 4.19  | 4.58  | 3.08  | 2.95  | 2.68  | 2.83  | 4.21  | 3.70  | 4.25  | 3.97 |
|                                                                                  | HDD10.0     | 1235   | 364   | 226   | 132   | 44    | 1     | 0     | 0     | 0     | 1     | 34    | 144   | 288  |
|                                                                                  | HDD18.3     | 3035   | 622   | 459   | 381   | 221   | 77    | 18    | 9     | 8     | 77    | 231   | 384   | 546  |
|                                                                                  | CDD10.0     | 1541   | 0     | 1     | 9     | 73    | 196   | 294   | 337   | 352   | 206   | 62    | 9     | 0    |
|                                                                                  | CDD18.3     | 299    | 0     | 0     | 0     | 1     | 13    | 63    | 88    | 101   | 33    | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 3094   | 0     | 0     | 2     | 49    | 188   | 626   | 838   | 1041  | 338   | 12    | 0     | 0    |
|                                                                                  | CDH26.7     | 1031   | 0     | 0     | 0     | 3     | 27    | 188   | 280   | 424   | 108   | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.4    | 2.5   | 2.7   | 2.8   | 2.9   | 2.5   | 2.1   | 2.3   | 1.9   | 2.3   | 2.2   | 2.1   | 2.8  |
| Precipitation                                                                    | PrecAvg     | 629    | 25    | 24    | 34    | 59    | 73    | 94    | 91    | 60    | 59    | 46    | 33    | 36   |
|                                                                                  | PrecMax     | 839    | 49    | 62    | 71    | 111   | 154   | 176   | 181   | 132   | 130   | 95    | 68    | 80   |
|                                                                                  | PrecMin     | 396    | 8     | 4     | 8     | 12    | 28    | 25    | 18    | 10    | 5     | 2     | 0     | 4    |
|                                                                                  | PrecStd     | 117    | 12    | 12    | 16    | 26    | 30    | 36    | 40    | 27    | 32    | 25    | 17    | 21   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 9.0   | 15.1  | 21.7  | 27.0  | 29.1  | 32.7  | 33.7  | 35.4  | 33.0  | 25.4  | 18.5  | 11.1 |
|                                                                                  |             | MCWB   | 6.8   | 9.7   | 11.6  | 15.7  | 18.0  | 21.7  | 20.6  | 20.8  | 20.5  | 16.4  | 13.3  | 8.1  |
|                                                                                  | 2%          | DB     | 7.1   | 12.4  | 18.1  | 24.7  | 27.3  | 30.6  | 31.9  | 32.9  | 29.8  | 22.8  | 15.3  | 8.9  |
|                                                                                  |             | MCWB   | 5.8   | 8.5   | 10.4  | 14.8  | 17.5  | 21.4  | 20.5  | 19.7  | 18.9  | 15.2  | 11.7  | 7.3  |
|                                                                                  | 5%          | DB     | 5.2   | 10.1  | 15.5  | 22.1  | 25.6  | 28.8  | 29.7  | 30.9  | 27.2  | 20.6  | 13.5  | 7.3  |
|                                                                                  |             | MCWB   | 4.2   | 7.2   | 9.1   | 13.8  | 17.0  | 20.2  | 20.0  | 19.6  | 17.6  | 14.4  | 10.8  | 6.1  |
|                                                                                  | 10%         | DB     | 3.8   | 8.3   | 12.9  | 19.8  | 23.6  | 27.2  | 27.7  | 28.9  | 24.9  | 18.0  | 11.7  | 5.6  |
|                                                                                  |             | MCWB   | 3.0   | 6.1   | 7.6   | 12.8  | 16.2  | 19.4  | 19.4  | 19.7  | 17.2  | 13.5  | 9.5   | 4.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 7.7   | 10.7  | 12.6  | 16.9  | 20.7  | 23.9  | 22.9  | 22.4  | 20.8  | 18.1  | 13.6  | 9.2  |
|                                                                                  |             | MCDB   | 8.8   | 14.7  | 20.6  | 24.0  | 26.3  | 29.6  | 29.1  | 30.3  | 31.2  | 22.8  | 18.1  | 10.0 |
|                                                                                  | 2%          | WB     | 6.1   | 9.0   | 10.8  | 15.7  | 19.0  | 22.4  | 21.9  | 21.6  | 19.6  | 16.1  | 12.3  | 7.4  |
|                                                                                  |             | MCDB   | 7.3   | 11.6  | 16.7  | 23.1  | 24.2  | 28.9  | 28.3  | 29.3  | 27.9  | 20.4  | 15.0  | 8.6  |
|                                                                                  | 5%          | WB     | 4.5   | 7.7   | 9.5   | 14.5  | 17.9  | 21.3  | 21.1  | 20.8  | 18.5  | 14.9  | 11.3  | 6.1  |
|                                                                                  |             | MCDB   | 5.2   | 9.9   | 14.5  | 21.4  | 23.1  | 27.3  | 27.5  | 28.2  | 25.0  | 19.5  | 13.6  | 7.2  |
|                                                                                  | 10%         | WB     | 3.2   | 6.3   | 8.3   | 13.3  | 16.9  | 20.3  | 20.3  | 20.2  | 17.6  | 13.7  | 9.9   | 4.9  |
|                                                                                  |             | MCDB   | 3.9   | 8.1   | 12.0  | 18.8  | 22.3  | 25.6  | 26.3  | 26.9  | 23.3  | 17.7  | 11.7  | 5.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.1   | 8.3   | 10.4  | 12.3  | 11.5  | 12.0  | 12.4  | 13.1  | 11.7  | 11.3  | 7.7   | 5.0  |
|                                                                                  |             | MCDBR  | 6.7   | 11.7  | 15.5  | 17.3  | 15.9  | 15.2  | 16.5  | 17.3  | 16.2  | 17.0  | 10.6  | 7.4  |
|                                                                                  |             | MCWBR  | 5.4   | 7.8   | 8.7   | 8.4   | 7.3   | 6.4   | 6.2   | 5.9   | 5.9   | 9.4   | 6.8   | 5.8  |
|                                                                                  | 5% WB       | MCDBR  | 6.8   | 10.5  | 13.5  | 15.7  | 13.1  | 13.3  | 13.9  | 14.5  | 14.3  | 15.0  | 9.8   | 7.2  |
|                                                                                  |             | MCWBR  | 5.5   | 7.2   | 8.0   | 7.7   | 6.6   | 6.5   | 5.8   | 5.6   | 5.5   | 8.5   | 6.5   | 5.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.323  | 0.354 | 0.401 | 0.463 | 0.447 | 0.440 | 0.460 | 0.471 | 0.423 | 0.381 | 0.351 | 0.323 |      |
|                                                                                  | taud        | 2.324  | 2.246 | 2.169 | 2.026 | 2.139 | 2.204 | 2.155 | 2.115 | 2.212 | 2.316 | 2.346 | 2.345 |      |
|                                                                                  | Ebn at Noon | 763    | 805   | 815   | 794   | 824   | 831   | 807   | 779   | 784   | 765   | 721   | 718   |      |
|                                                                                  | Edh at Noon | 78     | 102   | 128   | 161   | 149   | 140   | 145   | 146   | 121   | 95    | 77    | 69    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.27   | 2.06  | 3.19  | 4.33  | 5.18  | 5.78  | 5.68  | 5.21  | 3.71  | 2.58  | 1.50  | 1.02  |      |
|                                                                                  | RadStd      | 0.17   | 0.26  | 0.45  | 0.47  | 0.49  | 0.53  | 0.49  | 0.51  | 0.40  | 0.30  | 0.19  | 0.13  |      |

## Historical Trends

|                              |  | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) Regional (79 neighbors) |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                              |  | +0.69   | N/A | N/A    | +0.95   | N/A   | N/A   | -127        | -190    | +101    | +45     |

Nomenclature: See separate page